10. Applying Performance Optimization Techniques (Indexing, Subforms and Query Tuning) in Zoho Creator

Aim

To apply performance optimisation techniques in the lost and found application: field indexing, subform structuring and Deluge query tuning.

Overview

- Field indexing: an index on a field that is searched often avoids scanning the whole table. Zoho Creator does not expose manual database indexes; the related option is the No duplicate values setting on a field together with precise report criteria.
- Subforms: child rows are kept inside the parent form, which reduces the number of separate lookups.
- Query tuning: use precise criteria, fetch only the needed record, use count() instead of fetching to count, and keep fetches out of loops.

Techniques applied in the application

Technique	Where it is used	Status
Single criteria count instead of fetch and loop	DS-02: duplicate lost-item check	Applied
Fetch only the one linked record by ID	DS-07: status update of the found item	Applied
Subform for child rows	Claim_Requests: action date, action taken, remarks	Applied
No fetch inside a loop	Form workflow scripts DS-01 to DS-07	Applied
No duplicate values setting	A frequently searched field such as Email	Studied

Procedure

Step 1: Query tuning in the duplicate check (DS-02). Instead of fetching all lost records into a collection and looping over them, the script counts only the records that match the item name, date and location.

```
if(Lost_Items[Item_Name == input.Item_Name
   && Date_Lost == input.Date_Lost
   && Location_Lost == input.Location_Lost].count() > 0)
{
    alert "This lost item has already been reported.";
    cancel submit;
}
```

Step 2: Fetching a single record by ID (DS-07).

```
if(input.Claim_Status == "Approved")
{
    foundRec = Found_Items[ID == input.Found_Items];
    foundRec.Status = "Claimed";
}
```

Step 3: Subform structuring. The actions taken on a claim are stored as subform rows inside Claim_Requests, so a claim and its history open together (Fig. 10.1).

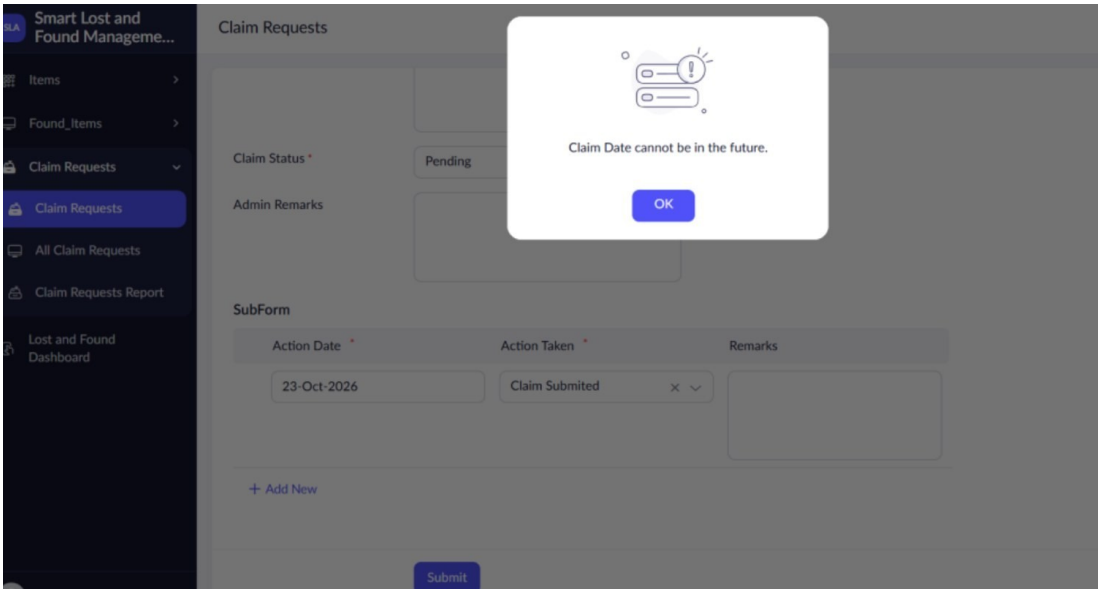


Fig. 10.1: Claim_Requests form with its subform of claim actions

Step 4: Timing. The time taken by a script can be read in the debugger with the script below. With about 33 test records the difference is too small to show; the benefit appears at larger data volumes.

```
t1 = zoho.currenttime.toLong();
c = Lost_Items[Item_Name == "keys"].count();
t2 = zoho.currenttime.toLong();
info (t2 - t1) + " ms";
```

Result

Query tuning (count instead of fetch and loop, fetch by ID) and subform structuring were applied in the application. With only about 33 test records the benefit would appear at larger data volumes.